

Sewon Kim

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Summary

Final-year Computer Science and Engineering student building AI and data systems across international organizations, research labs, and field deployments. Currently supporting knowledge discovery and AI enrichment at the FAO Office of Innovation; previously delivered a cultural-heritage inspection system used by the Korea Heritage Service and co-authored an IoT systems preprint.

Technical Skills

Programming	Python, C/C++, SQL, Dart, Java
ML and Data	PyTorch, scikit-learn, Hugging Face, LangChain, OpenCV; RAG, knowledge graphs, computer vision, NLP, and time-series modeling
Systems and Tools	FastAPI, Flask, Firebase, Docker, AWS, GCP, Git, Linux, MQTT, Flutter

Experience

Innovation Accelerator Intern FAO Office of Innovation	Jun 2026--Present Rome, Italy
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- Analyze user behavior and knowledge-discovery flows for the ATIO Knowledge Base, supporting UX and data-visualization recommendations across **5,744** agricultural technology and innovation records.
- Evaluate and refine AI-enrichment prompts used to classify records and improve metadata quality and consistency.
- Collaborate with UX, Drupal, taxonomy, and data-harvesting workstreams in a multicultural, cross-functional team.

Undergraduate Researcher Natural Language Learning Lab, Jeonbuk National University	Feb 2025--Feb 2026 Jeonju, South Korea
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- Developed knowledge graph-enhanced RAG architectures in PyTorch and LangChain, and ran ablation studies on retrieval accuracy, latency, and context coherence for multi-hop question answering.
- Led the architecture and implementation of a DETR-based cultural-heritage damage-inspection system, achieving an **F1 score of 0.89** and resulting in operational use, a registered software copyright, and a patent filing.

International Research Scholar, IITP-Funded Korean Software Square Summer Program, Purdue University	Jun 2025--Aug 2025 West Lafayette, IN, USA
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- Contributed to the design, field deployment, and evaluation of WaggleNet, a LoRa-MQTT beehive-monitoring system that achieved **100% packet delivery at 110 m, 3.8 s** mean latency, and over **95% uptime** during a two-month trial.
- Co-authored the resulting arXiv preprint and helped validate a **\$25--30 per-node** design estimated at **70--85% lower cost** than commercial alternatives.

Experience

Teaching Assistant, Algorithms and Linux Programming

Jeonbuk National University

Sep 2025--Dec 2025

Jeonju, South Korea

- Mentored **80+ students** in Algorithms and **50+ students** in Linux Programming; built grading scripts and rubrics to standardize assignment and exam evaluation.

Publications and Intellectual Property

Preprint M. Jeon, J. Kim, S. Kim, S. Park, B. Zhang, and A. H. Smith. "WaggleNet: A LoRa and MQTT-Based Monitoring System for Internal and External Beehive Conditions." arXiv:2512.07408, 2025.

Registered software copyright "Integrated Survey Registration and Damage Diagnosis Management App," Korea Copyright Commission, Reg. No. C-2025-044615, 2025.

Patent filed, pending Integrated survey and damage-diagnosis workflow for cultural-heritage structures; jointly filed by Jeonbuk National University and the Korea Heritage Service, 2025.

Education

B.S. in Computer Science and Engineering

Jeonbuk National University | GPA: 4.09/4.5 (3.64/4.0)

Mar 2022--Aug 2026 (Expected)

Jeonju, South Korea

- Full-tuition Undergraduate Merit Fellowship; admitted with Highest Distinction as the top-ranked entrant in the College of Engineering.

Exchange Student, Computer Science

Universiti Malaya

Feb 2023--Aug 2023

Kuala Lumpur, Malaysia

Selected Honors

- President's Award for Excellence in Core Competencies, highest honor, Jeonbuk National University (2025).
- President's Award, Keun-Saram Black Belt Scholarship, competitive merit scholarship of \$3,400 (2025).
- Excellence Award, Capstone Design Competition, two-time recipient (2024--2025).
- Commendation, Big Data Supporters Program, Ministry of Science and ICT, Republic of Korea (2024).

Additional Information

Languages

Korean (native), English (full professional proficiency), Japanese (upper-intermediate), Spanish (intermediate)

Certification

Data Architecture Semi-Professional (DAsP), Korea Data Agency, 2025

Leadership

Lead Student Representative, Big Data Supporters (2024--2025); Student Representative, Honors Residential College (2022--Present)